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Near-threshold fatigue crack propagation and internal cracks in steel

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Abstract

The existence or non-existence of fatigue crack growth thresholds is of practical importance and scientific interest, as is knowledge of the stresses enabling crack formation at the surface and the interior of a specimen. Near-threshold measurements have been performed on 12% chromium steels with an ultrasound testing technique at different mean loads in ambient air and vacuum at very high numbers of cycles. Microstructural features and fracture surfaces were investigated in an SEM. The fracture appearance after cycling in vacuum in the near threshold regime is compared with that of the fine granular and smooth areas of internal "fish-eyes".

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Keywords: Fatigue Crack Growth; Very High Cycle Fatigue; Stress Intensity Thresholds; 12% Chromium Steel; Ultrasonic Resonance Testing; Fish-eye Fracture, Internal Crack Initiation

1. Introduction

Very long lifetimes of moving machine parts became more and more important during the last decades. Therefore, the interest in crack growth rates near threshold in the very high cycle fatigue (VHCF) regime is increasing. Since the investigated material finds use in turbine construction, mechanical engineering and naval construction, the acquired data are of practical relevance.

For the existence of a threshold value for crack propagation (and fatigue limit) the influence of environment plays an important role. Moisture in air, for example, can act like a corrosive medium at the crack tip and embrittlement may occur by absorbed hydrogen which might induce sub-threshold fatigue crack propagation. Oxides and other debris, in contrast, may enhance crack closure. To investigate these effects, measurements not only in corrosive but also in inert environment (e.g. vacuum) have to be performed for the purpose of comparison.

Measurements in the VHCF regime, which are necessary to observe fatigue crack growth in the threshold fatigue regime, are very time-consuming with conventional technologies (which operate at frequencies of some Hz up to a maximum of 1000 Hz). For this reason, investigations are nowadays performed with the ultrasound fatigue testing technique with high accuracy, which is an efficient method to reduce testing times.

In the VHCF regime, several materials show internal crack initiation instead of cracks starting from the surface.

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This leads to a dual slope S-N curve [1-5]. Sub-surface cracks start at inclusions (or cavities) and the fracture surface may show the well-known “fish-eyes”. The latter show a fine granular looking area surrounding the inclusion, if failure occurs at low stress amplitude. This area is named ODA (optically dark area) by Murakami et al. [6], FGA (fine granular area) by Sakai et al. [7], and GBF (granular bright facet) by Shiozawa et al. [8]. It will be referred to as FGA in the present work. Since most of the lifetime is spent forming the FGA, this kind of fracture behaviour is a matter of particular interest. The FGA at an inclusion is formed after about 5-10% of fatigue life and grows for 90% of lifetime [9,10], which means that the crack growth rate is far below one Burgers vector per cycle [8-11]. The size of the FGA increases with decreasing stress amplitude and increasing fatigue life [6,12]. Inclusions leading to an FGA cause stress intensity factors ΔK_{inc} which are much lower than threshold values ΔK_{th} for long crack propagation if calculated by using Murakami's $\sqrt{\text{area}}$ model for subsurface fracture mode [12]. Replacing ΔK_{inc} by ΔK_{FGA} , which takes into account the size of FGA (including the size of inclusion), shows that ΔK_{FGA} is in the range of ΔK_{th} after all [6,11,14].

There are different models suggesting mechanisms for FGA formation: Murakami et al. assume hydrogen embrittlement caused by hydrogen trapped around the inclusion [6], Sakai et al. suppose the formation of small subgrains which are separated with time [7], and Shiozawa et al. propose multiple microcracks initiated by decohesion of spherical carbides in the vicinity of an inclusion [8].

In the present paper, fatigue crack growth rates were measured for 12% chromium steel in the near-threshold regime. The resulting fracture surfaces characterizing mean crack growth rates below one atomic distance per cycle are compared with those of specimens where internal (fish-eye) fractures formed. Emphasis is laid on the FGA and surrounding smooth area (SA). The present studies aim to gain a deeper insight in the mechanisms of internal cracking and VHCF.

2. Material and experimental setup

2.1. Testing Material

Fatigue crack growth measurements were performed with martensitic 12% chromium steel AISI410 (material number 1.4006, DIN X12Cr13, JIS SUS410M). The chemical composition can be seen in Table 1. The material was heat-treated in the following way: 913°C (2 h/oil quenched) + 649°C (4 h/air cooled) + 621°C (4 h/air cooled). The mechanical properties are shown in Table 2 (material A). Tubular specimens (shown in Fig. 1a) were machined, ground and polished with abrasive paper (up to grade #4000) to get a mirror finish. To eliminate residual stresses, which influence the crack growth behaviour, the specimens were stress-relieved by annealing at 600°C for two hours and slowly cooling afterwards (approx. 100°C per hour). A hole was drilled and a notch introduced by spark erosion (with mineral oil as protective liquid) into the wall centre to cause crack initiation there (length of the notch including drilling: approx. 4 mm). The cracks grew along the circumference of the tube in the plane of maximum normal stress. Strain gauges were attached to the specimen to calibrate the loading amplitudes.

In order to obtain internal crack initiation, a similar material (AISI420) was tempered differently: 980°C (1 h/oil quenched) + 300°C (2 h/air cooled). The mechanical properties are shown in Table 2 (material B). Specimens were machined and ground with abrasive paper (up to grade #500) before heat-treatment and ground to grade #1000 afterwards.

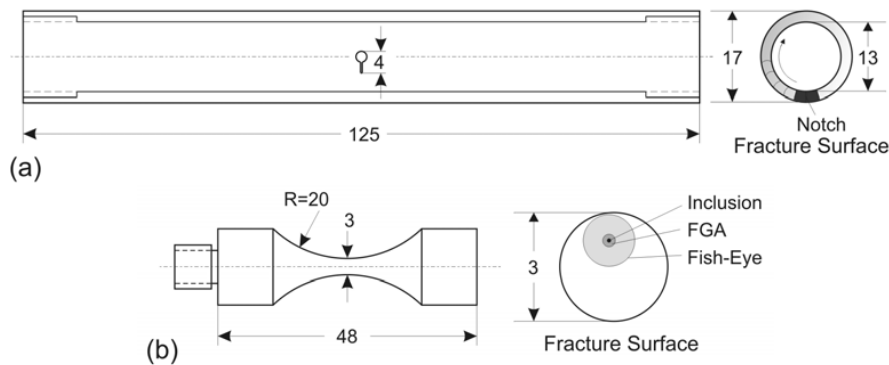


Fig. 1. (a) Specimen shape (type A) for fatigue crack growth measurements (arrow in fracture surface indicates crack growth direction) and (b) type B specimen shape for internal crack initiation.

Table 1. Chemical composition of the testing material A and B (in % weight)

Material	C	Cr	Mn	Si	Ni	Mo
A	0.130	11.88	0.41	0.22	0.33	0.18
B	0.20	12.50	0.40	0.40	-	-

Table 2. Mechanical properties of the testing material from material A (tempered at 621°C) and material B (tempered at 300°C)

Material	Yield Strength (MPa)	Tensile Strength (MPa)	Elongation (%)	Area Reduction (%)	Hardness HV	Modulus (MPa)	Grain Size (ASTM #)
A	610	800	22,5	67	260	215	9.3
B	1250	1500	-	-	460	215	-

2.2. Testing method

Fatigue crack growth rates in the near-threshold regime were measured at different load ratios ($R = +0.05$, $R = +0.5$ and $R = +0.8$) and in different environments (ambient air at 22°C with a relative humidity (RH) of 50% and vacuum of approx. 10^{-3} Pa) using ultrasonic fatigue testing equipment. This technique works by stimulating specimens to resonance vibrations at about 20 kHz cycling frequency, which allows a number of cycles of more than 10^9 within one day. A closed-loop control guarantees that the pre-selected and the actual vibration amplitude coincide with an accuracy of typically 99%. To avoid a rise of temperature during testing at 20 kHz, the specimens were loaded by series of pulses and pauses. The pulse length was 100 ms (2000 cycles) and pause lengths were selected according to the applied amplitude (the higher the amplitude the longer the pause). With these pauses and an additional fan the temperature of the specimens was kept below 30°C. Ultrasonic fatigue testing works at fully reversed tension-compression loading ($R = -1$), if one end is allowed to vibrate freely. R-values different from -1 can be obtained by attaching the specimen's end to the coupling piece of a mechanical or a servohydraulic testing equipment. For a detailed description of ultrasonic fatigue and fatigue crack growth see [4,15].

Fatigue crack growth was measured at the surface of the specimen using an optical system including a CCD-camera. The magnification at the monitor was 220-fold with an optical resolution of 10 μm .

The measurements were started at a relatively high ΔK value (Paris regime, crack growth rates between 10^{-8} and

10^{-9} m/cycle) to introduce a starter crack. Then the stress amplitude was decreased in steps of 7 to 8 percent. At each stress level the sample was stressed until the crack obtained a length of 15 - 50 mm on the monitor (70 - 230 μm real length). The stress amplitude was lowered until the crack did not grow within at least $1 \cdot 10^8$ cycles. In the ($\Delta a/\Delta N$ vs. ΔK) curves, these points are marked with an arrow (a crack length of 10 μm was assumed as this is approximately the optical resolution length). Then the amplitude was increased to start crack growth again. This procedure was repeated until the total crack length (including notch and drilling) was approx. 11 mm, which is the upper limit of measurement. The specimens were fractured at high cyclic stresses after the test. Fracture surfaces of all specimens were investigated with optical and scanning electron microscope (SEM). Optical microscope inspection allowed controlling the validity of measurement, i.e. whether the crack grew uniformly or not. SEM observation was used for characterizing the fracture mode.

3. Results and Discussion

3.1. Fatigue crack growth experiments in vacuum

Fatigue crack growth data of AISI410 determined at a cyclic frequency of 19 kHz under different R-ratios in vacuum are shown in Fig. 2a. Arrows indicate where no crack growth could be detected within at least $1 \cdot 10^8$ cycles. Three specimens were used to measure these curves, and with one specimen, experiments at different R-ratios can be performed. Scattering of the results was very low.

As expected, the cyclic stress intensity threshold values increase with decreasing R-ratio if plotted vs. ΔK . Crack propagation still takes place at rates as low as $5 \cdot 10^{-13}$ meters per cycle (m/cycle) which is a factor of 200 below a mean crack propagation rate of one lattice space per cycle.

3.2. Fatigue crack growth experiments in ambient air

Fig. 2b shows the fatigue crack growth rates for ambient air. Again, three specimens were used for determination of the curves. The scatter is somewhat larger than in vacuum, the reason probably being environmental effects.

The measurements yield the following results:

The lowest crack propagation rates are between $2 \cdot 10^{-11}$ m/cycle ($R = +0.05$) and $6 \cdot 10^{-12}$ m/cycle ($R = +0.5$), which is still a factor of 5-17 below a mean crack propagation rate of one lattice space per cycle. The curves for different R-ratios converge in the range of higher ΔK values (begin of Paris regime). For $R = +0.8$ and $R = +0.5$, the crack growth rates even become identical above $5 \cdot 10^{-10}$ m/cycle, approaching the curve for $R = +0.05$.

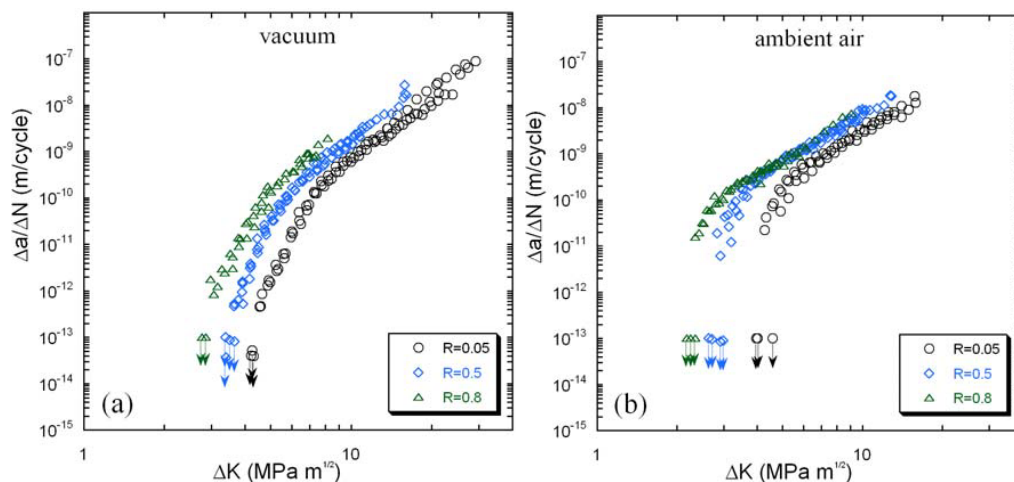


Fig. 2. Fatigue crack growth in (a) vacuum (approx. 10-3 Pa) and (b) ambient air (22°C and RH = 50%).

3.3. Comparison of fatigue crack growth experiments in vacuum and ambient air

Comparison of Figs. 2a and 2b shows that higher crack growth rates and lower stress intensity thresholds can be observed in ambient air as opposed to vacuum. This result is not unexpected, since water vapour of ambient air moisture acts as corrosive environment and may lead to hydrogen embrittlement and other crack growth enhancing processes.

The threshold stress intensity factors ΔK_{th} in vacuum and air differ considerably less (5-30%) than the K-values for crack growth rates of, for example, 10^{-10} m/cycle (55-70%). The values from Fig. 2a and 2b are clarified in Fig. 3, where $\Delta a/\Delta N$ at 10^{-10} m/cycle (which is used as threshold crack growth rate quite often) and the mean values of the actually measured limiting crack propagation rates in vacuum and air were marked by horizontal lines for comparison (the corresponding values for stress intensities at these crack growth rates are added in brackets in Table 3). As mentioned above, fatigue crack propagation in vacuum still takes place at a rate of $5 \cdot 10^{-13}$ m/cycle, while the lowest rate in ambient air is $6 \cdot 10^{-12}$ m/cycle. This is probably the reason for the surprising result that the threshold stress intensity factors measured in inert environment are not so different from those in air, because crack propagation in vacuum still takes place at ten times lower crack growth rates than observed in ambient air. The reason for the thresholds at higher crack growth rates in ambient air could be oxide-induced crack closure, which does not allow crack propagation much below 10^{-11} m/cycle.

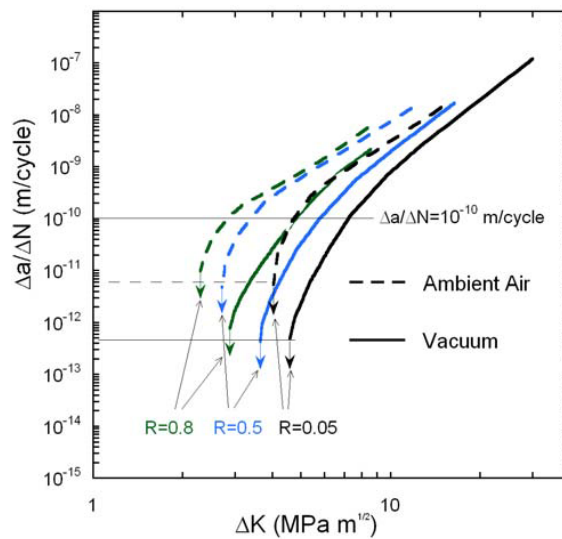


Fig. 3. Comparison of fatigue crack growth in vacuum of approx. 10^{-3} Pa (solid lines) and air at 22°C with $\text{RH} = 50\%$ (broken lines). Extracted from Fig. 2a and 2b.

Table 3. Comparison of threshold stress intensity factor ΔK_{th} (ΔK at 10^{-10} m/cycle in brackets) in vacuum and ambient air at different R-ratios ($\text{MPa} \sqrt{\text{m}}$)

Environment	R = +0.05	R = +0.5	R = +0.8
Vacuum	4.23 (7.3)	3.39 (5.7)	2.77 (4.8)
Ambient Air	3.98 (4.7)	2.64 (3.4)	2.18 (2.8)

3.4. Fracture morphology of long surface cracks and fish-eye fractures

The fracture surfaces were observed in the SEM. In material A, cracks started from the notches and propagated perpendicular to the load axis in circumferential direction (see Fig. 1). The crack growth direction in Fig. 4 is from bottom to top. In Fig. 4a, the border between near threshold crack growth (lower part of image) and a higher crack growth rate after crack arrest (upper part of image) in vacuum is shown. The same can be seen in Fig. 4b for ambient air. In the near threshold regime mainly transcrystalline fracturing with a relatively smooth surface is visible for vacuum as well as for an ambient air environment. With increasing stress intensity, the fracture appearance changes to a rougher and more ductile fracture mode. In humid air, the fracture surfaces are generally rougher, and some intergranular and crystallographic facets become visible in addition.

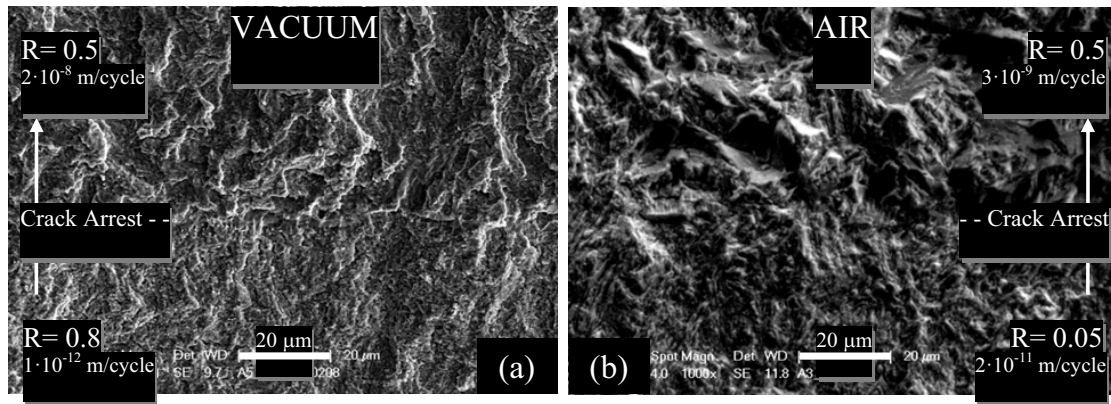


Fig. 4. Fracture surface of a long crack formed by near threshold cycling (lower part) and by higher crack growth rates (upper part). Crack arrest is visible along the horizontal borderline. The crack growth direction was from bottom upwards. (a) vacuum: $\Delta a/\Delta N = 1 \cdot 10^{-12}$ m/cycle ($R = +0.8$) in the lower part and $\Delta a/\Delta N = 2 \cdot 10^{-8}$ m/cycle ($R = +0.5$) in the upper part of image; (b) air: $\Delta a/\Delta N = 2 \cdot 10^{-11}$ m/cycle ($R = +0.05$) in the lower part and $\Delta a/\Delta N = 3 \cdot 10^{-9}$ m/cycle ($R = +0.5$) in the upper part of the image.

To obtain information on the crack growth rates within the three fish-eye zones (FGA, SA and Paris-regime area) in a type B specimen (solid fatigue specimen, Fig. 1b), the fracture surfaces of the type A (tubular, Fig. 1a) specimens which had been used to determine the ($\Delta a/\Delta N$ vs. ΔK) curves were analysed. Since it was assumed that vacuum is prevailing for the interior cracks in type B specimens, the type A specimens were likewise fatigue loaded in vacuum [16]. Thus, the crack growth rates at specified areas of these specimens were known. Comparing their fractographic features with the fish-eye features allowed to estimate the local crack growth rates there. Fig. 5a is a higher magnification of the lower part of Fig. 4a, where the crack growth rate was $1 \cdot 10^{-12}$ m/cycle. Fig. 5b is a magnification of the fracture surface of an unnotched, polished specimen, which had been fatigue loaded with $\Delta \sigma/2 = 677$ MPa. Fracturing occurred after $8.5 \cdot 10^8$ cycles due to internal crack initiation with a fish-eye fracture. At this magnification, the similarity of the fracture surface morphology and roughness of Fig. 5a and 5b becomes obvious. This means that the crack growth rate within the “fine granular area” [7] (or “optical dark area” [6] or “granular bright facet” [8]) in the vicinity of an inclusion must have been close to $1 \cdot 10^{-12}$ m/cycle, a value which has also been reported by Tanaka and Akiniwa [9].

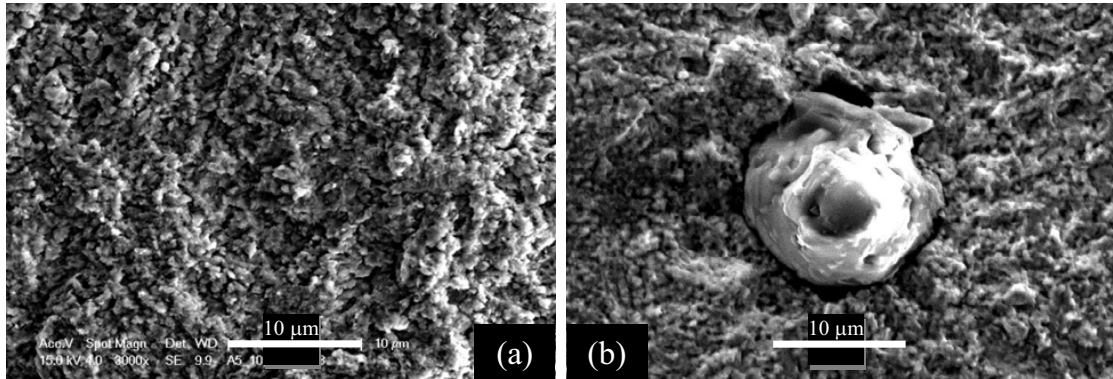


Fig. 5. (a) Fracture surface of a long crack formed by near threshold cycling with $\Delta a/\Delta N = 1 \cdot 10^{-12}$ m/cycle ($R = +0.8$); (b) fracture surface of a type B specimen; inclusion in the centre of a fish-eye with FGA ($\Delta\sigma/2 = 677$ MPa, $N_f = 8.5 \cdot 10^8$ cycles). Both images with identical magnification.

Another interesting feature of the fish-eye fracture is the smooth area (SA) surrounding the FGA (circle with broken line in Fig. 6b), which is even smoother than the FGA (circle with solid line in Fig. 6b) where the fatigue crack growth rate was found to be close to $1 \cdot 10^{-12}$ m/cycle. The former area may be considered as the stage II fatigue crack growth area, following stage I cracking in the FGA according to Tanaka and Akiniwa [9]. In this SA, fatigue crack growth rates are probably somewhat higher, i.e. in the range of $1 \cdot 10^{-11}$ m/cycle, as may be assumed after comparison with Fig. 6a, which stems from measurements with type A specimens. Tanaka and Akiniwa modelled internal mode I and mode II crack propagation and found that the crack growth rates within the FGA as well as the SA must be lower than those for usual long cracks.

The upper part of Fig. 6a characterizes the fatigue crack growth features above $\Delta a/\Delta N = 1 \cdot 10^{-9}$ m/cycle (Paris regime) in a type A specimen, and the similarity with the outer part of the fish-eye fracture, neighbouring the smooth part in Fig. 6b is obvious. Summarizing these observations, one can assume that interior cracks starting with a growth rate close to $1 \cdot 10^{-12}$ m/cycle at an inclusion become faster when propagating radially outwards from a fish-eye with $\Delta a/\Delta N \leq 1 \cdot 10^{-11}$ m/cycle in the SA and reach crack growth rates above $\sim 10^{-9}$ m/cycle in the third region of the fish eye, before final fracturing takes place.

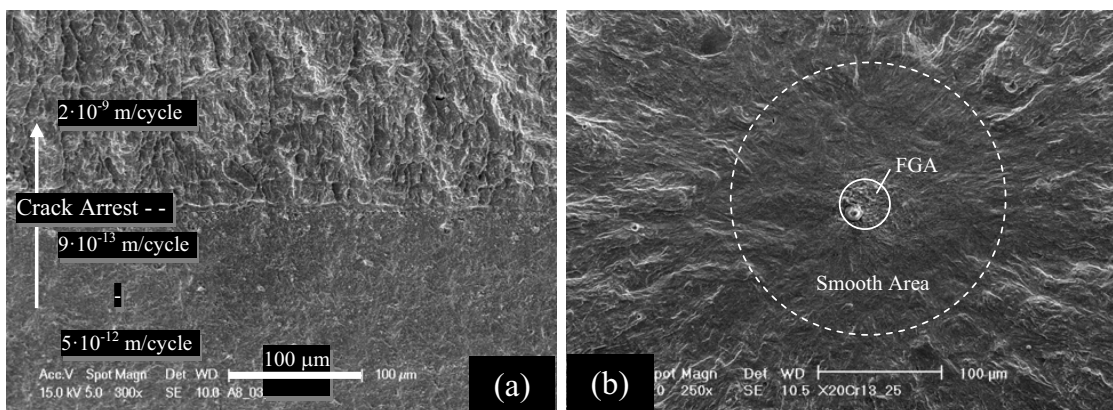


Fig. 6. (a) Fracture surface of 12% Cr steel type a formed by fatigue crack growth near threshold cycling (lower part) and by higher crack growth rates (upper part) in vacuum: $\Delta a/\Delta N \approx 10^{-12}$ m/cycle ($R = +0.05$) in lower part and $\Delta a/\Delta N \approx 10^{-9}$ m/cycle ($R = -1$) in upper part of image. Crack arrest was along the horizontal borderline. Crack growth direction was from bottom upwards; (b) Internal fracture surface of 12% Cr steel type B. Fish-eye with FGA and inclusion (solid line) and SA (broken line) ($\Delta\sigma/2 = 677$ MPa, $N_f = 8.5 \cdot 10^8$ cycles).

4. Conclusions

Fatigue crack growth rates of martensitic 12% chromium steel AISI410 are determined in the near threshold regime at different load ratios ($R = +0.05$, $R = +0.5$ and $R = +0.8$) and in different environments (ambient air at 22°C and RH = 50% and vacuum of approx. 10^{-3} Pa) using ultrasonic fatigue testing equipment at a cyclic frequency of about 20 kHz. The resulting fracture surfaces are compared with those of other specimens (similar material (AISI420) but tempered differently) where internal (fish-eye) fractures occurred. Emphasis is laid on researching the fine granular area (FGA) in the neighbourhood of an inclusion and the surrounding smooth area (SA).

1. Higher crack growth rates and lower stress intensity thresholds are observed in ambient air than in vacuum owing to the corrosive influences by air moisture.
2. Fatigue crack propagation in vacuum still takes place at a rate of $5 \cdot 10^{-13}$ m/cycle, which is several decades below the theoretical value of one lattice space per cycle. The lowest rate in ambient air is $6 \cdot 10^{-12}$ m/cycle.
3. The threshold stress intensity factors ΔK_{th} in vacuum and air differ considerably less (5-30%) than the ΔK values for higher crack growth rates (at 10^{-10} m/cycle: 55-70%).
4. SEM observation of fracture surfaces shows mainly transcrystalline fracturing with a relatively smooth surface in the near threshold regime. For higher stress intensities, the fracture appearance changes to a rougher and more ductile fracture mode. In humid air, the fracture surfaces are generally rougher, and some intergranular and crystallographic fracture features become visible in addition.
5. Comparison of fracture surfaces of fatigue crack growth specimens tested in vacuum and specimens where fish-eye fractures formed shows similar features in the VHCF range. Thus, fractographic features which are identified as typical for defined crack growth rates in vacuum allow estimating the crack growth rates in the different fish-eye areas.

Following values could be allocated to the different areas:

- Fine Granular Area (FGA): $\sim 10^{-12}$ m/cycle
- Smooth Area (SA) around FGA: $\leq 10^{-11}$ m/cycle
- Area around SA (~Paris Regime): $\geq 10^{-9}$ m/cycle.

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